Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Empathy Coders
Project / Solution Name	SmartOrder Hub
Name (Leader)	YANG JIAHUI
Email & Phone (Leader)	jacky666yang@gmail.com / +60 1123591658
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://smart-order-hub-1.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Jacky-Godlike/Empathy-Coders

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The React/TypeScript single-page app is served by a FastAPI API, which also serves the production build on Replit. The API exposes workspace, message, order, catalog, risk, channel, email and activity endpoints. Inbound messages arrive through demo channel adapters, manual or CSV import, or a read-only IMAP connector. The extraction layer calls an optional AI provider (DeepSeek V4.1 Flash) and validates the response; if the provider is unavailable, a deterministic parser handles the message. The rules engine calculates stock, supplier lead time, revenue, cost, profit, margin, completeness and anomaly, then produces a weighted risk score. The owner approves, rejects or requests information; the API writes the order record, purchase draft and activity event. A mock OrbyteAdapter marks the integration seam; no live Orbyte API is called. Credentials are environment variables or Replit Secrets, and all demo data is synthetic and stored in a file-locked JSON document.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11	Backend runtime	PSF-2.0
TypeScript 7 / JavaScript	Frontend application code	Apache-2.0
React 19.3; React DOM 19.3; Vite 8.3; plugin-react 6.1; Tailwind CSS 4.3; lucide-react 0.546; PapaParse 5.7	UI, build, styling, icons, CSV parsing	MIT / ISC
FastAPI 0.128; Pydantic 2.13; Uvicorn 0.39; httpx 0.28; python-dotenv 1.2; openpyxl 3.1; pytest 8.4; google-genai 1.47 (optional)	API, validation, server, HTTP, env, Excel, tests, optional Gemini client	MIT / BSD-3-Clause / Apache-2.0
Node.js 22 / npm	Frontend toolchain	MIT / Artistic-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek Chat Completions API (V4.1 Flash)	Optional structured extraction and short risk brief; rules-only mode available	Free / paid
IMAP mailbox (Gmail / Outlook / provider)	Optional read-only email ingestion from a dedicated test mailbox	Free provider account

Note: WhatsApp and WeChat are not connected; their APIs require business accounts. Channel messages in the demo are simulated.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (including Replit Secrets)	Builds React, runs FastAPI on port 3000 and serves the public prototype; Secrets stores AI and mailbox credentials	SDC-provided Replit Core; Replit Cloud
GitHub	Source control, documentation and judge access	Free public repository
Cloud database / auth	None; file-locked JSON store and simulated RBAC	N/A
RunPod	Not used; no GPU workload in this project	N/A

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek V4.1 Flash	DeepSeek	Pre-trained; optional runtime	Provider API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic SME order dataset	Team-generated; included in repository	Synthetic / private	Team-owned	No
No third-party dataset	N/A	None	N/A	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
DeepSeek V4.1 Flash	Writing code and integrating team members' collaborative code	Major
Anthropic Claude Code	Code generation, review, testing and documentation	Moderate
OpenAI ChatGPT-6	Ideation and code/content review	Moderate
Google Gemini	Ideation and code/content review	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code was uploaded to Replit and deployed. We manually reviewed each feature in the deployed app to confirm it worked correctly. We checked that no API keys, passwords or other secrets were committed to the repository; credentials were stored only in Replit Secrets. The team reviewed every change, ran the automated tests, TypeScript type-check and production build, and kept human approval for consequential actions.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud platform / Replit Core credits	Builds the React frontend, runs FastAPI and serves the public prototype; Replit Secrets stores credentials.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is the official SDC-provided cloud platform and hosts the judged live prototype. The core logic is portable to another Python/Node host, but Replit provides the required deployment environment, public URL and Secrets storage.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

All customer, order, product, supplier and stock records are synthetic. The demo does not collect real customer, payment or statutory data. Messages pasted or imported by a judge are stored in the local JSON demo store and can be reset. If a test mailbox is connected, the app uses read-only IMAP, fetches message bodies, and never sends, deletes or moves mail; the password is held in server session memory or Replit Secrets, never written to the repository or returned to the browser. When AI mode is enabled, synthetic message text may be sent to the configured AI provider. The public demo is not multi-tenant and has no production authentication; it must not be used with real customer data.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI extraction can misread ambiguous or multilingual messages, and provider output may vary or be rate-limited. Mitigation: a deterministic rule parser, strict validation, visible confidence, a rules-only mode and mandatory human approval. The risk score is an explainable heuristic, not a certified credit or safety model. The demo uses a file-locked JSON store, not a production database, and simulated role-based access control; a public deployment should add real authentication, rate limiting and a database. The Orbyte integration is a mock because live platform access was not available. Email depends on provider IMAP and app-password policy. No real payments, purchase orders or statutory filings are made. The workflow still requires pilot validation with SMEs.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: 

Date: 25 October 2026